Software Requirements Specification

for

Academic & Schedule Management System

Version 0.1

Prepared by

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
0.1	Samith Reddy Abhishek Sardar Deepak Rajeshwari K Sana Uday Wankhede Tandel Darshan Navinkumar Bharath Chandra Darla Dharavath Shailaja Siddharth S Godolkar	First Draft	23/01/2026

1 Introduction

1.1 Product Scope

*The **Academic & Schedule Management System (ASMS)** is a comprehensive mobile and web application designed to serve as a personal academic assistant for students. The system aims to optimize student productivity by providing a centralized platform to organize daily schedules, track personal attendance records, manage academic tasks, and analyze performance trends, ensuring students stay on top of their coursework independently.*

Main Goals and the benefits of this product :

- *To help students to balance and plan their Academics.*
- *To help students to balance their workload by allowing them to prioritize assignments and link personal to-do lists directly with the academic calendar.*
- *To check the real time performance in the course attendance and academic progress.*

1.2 Intended Audience and Document Overview

Intended viewers :

As we start the design, developers will use this document to communicate and proceed. Then at the time of testing, testers would use this document. It can then be used to form a user manual using some subset of SRS.

Sections and Organization :

The SRS starts with sections that describe its importance and how to use it, then it moves forward with the overview and functionality of the product. Then it covers a guide towards the interfaces which make it easier for a user to understand how they need to interact with the product, covering some use cases to wind up.

Sequence for Important Sections :

- *A Developer or a Tester can start reading the document with focus on overview (2.1) functionality (2.2), Design and Implementation (2.3), interfaces (3.1) and Functional requirements (3.2).*
- *A user should focus on the scope (1.1), overview (2.1) and functionality (2.2) and then the specific requirements, going through sections 3.1.1, 3.1.3 and 3.2 .*

1.3 Definitions, Acronyms and Abbreviations

- ***SRS :*** *Software Requirements Specification*
- ***OTP :*** *One-Time Password*
- ***Admin :*** *System Administrator*
- ***UI :*** *User Interface*
- ***ASMS :*** *Academics and Schedule Management System*
- ***API :*** *Application Programming Interface*

1.4 Document Conventions

Formatting Conventions :

- *This document is written with an Arial font of size 12 with single spacing and 1-inch margins.*
- *Words highlighted with the bold in the same font space represent terms whose explanations are either given in footnotes or separately in the same section.*
- *Underline and bold has been used for headings in the subsections.*
- *Bullet point ordering has been used as a listing typesetting tool.*

1.5 References and Acknowledgments

We'd like to acknowledge the help of our TA, Mr. Aditya Ojha, for their valuable input in the creation of this document. We also would like to thank Prof. Indranil Saha for providing the SRS template and teaching the concepts.

2 Overall Description

2.1 Product Overview

The Academic & Schedule Management System (ASMS) is a new and self-contained software system developed to help students manage academic activities and personal schedules in one place. It is not a follow-on or extension of any existing system, but an independent application designed specifically for student use.

The system is intended to reduce the need for students to rely on multiple separate tools such as notebooks, calendars, and messaging applications to keep track of academic work. ASMS provides a centralized platform where students can interact with different academic management features through a single user interface.

The system interacts mainly with students as end users and a backend data storage system where academic and scheduling data is maintained. Students access the system through common user devices such as mobile phones or laptops using an internet connection. ASMS does not depend on any external academic systems and functions independently.

2.2 Product Functionality

The Major Functionalities of the application include:

- User Account creation and secure login using roll number, phone number or email id with OTP verification.
- Storage and management of student academic data
- Management of course-related academic resources
- Attendance tracking for different courses
- Maintenance of academic calendar including quizzes, exam dates, venues and assignment submission deadlines
- Display of weekly timetables and academic events
- Task and To-Do list management of daily, weekly, and long-terms tasks
- Tasks prioritizing and remainder notifications before deadlines
- Viewing results of Quizzes and Exams
- Analysis of Academic performance using graphical representation

2.3 Design and Implementation Constraints

Hardware Limitations :
We need a computer/mobile and an internet connection for this application to build and execute.

Software Specifications :
We need an internet browser like Google Chrome, Brave, Mozilla Firefox etc. and an operating system that runs on 64 bit architecture for this application and its dependencies to run smoothly.

2.4 Assumptions and Dependencies

Assumptions

- Students have access to smartphones or computers with internet connectivity.
- Users will provide accurate academic and schedule-related information.
- OTP services will function correctly for user authentication.
- Students regularly update their tasks and academic data.

Dependencies

- Dependence on internet connectivity for login, synchronization and notifications.
- Dependence on third-party services for OTP verification.
- Dependence on external libraries or frameworks for data visualization.(graphs and analysis)
- Dependence on database systems for secure storage and retrieval of user data.

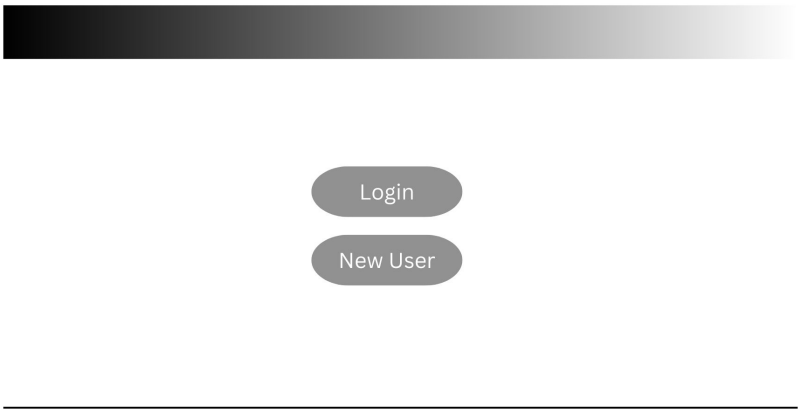
3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

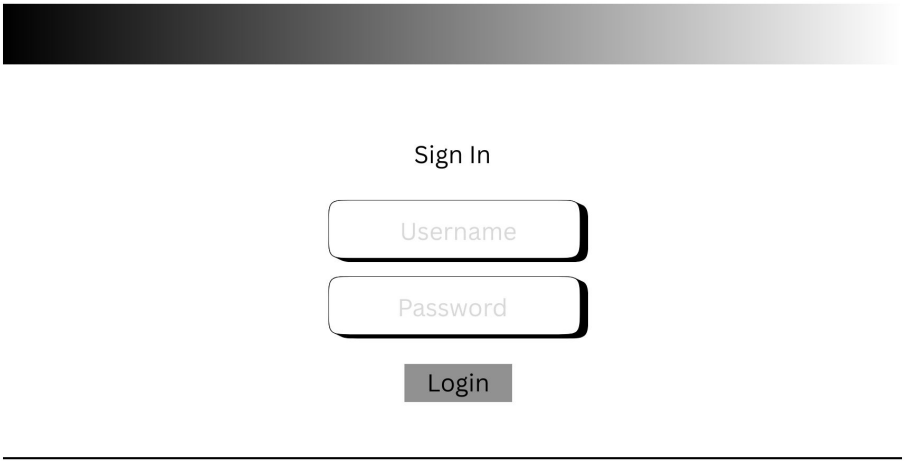
1. Homepage :

The homepage will contain the clean UI. It consists of two buttons: 'sign in' and 'new user'. The new users can register by clicking the button "New User". The registered user can login by the login button.



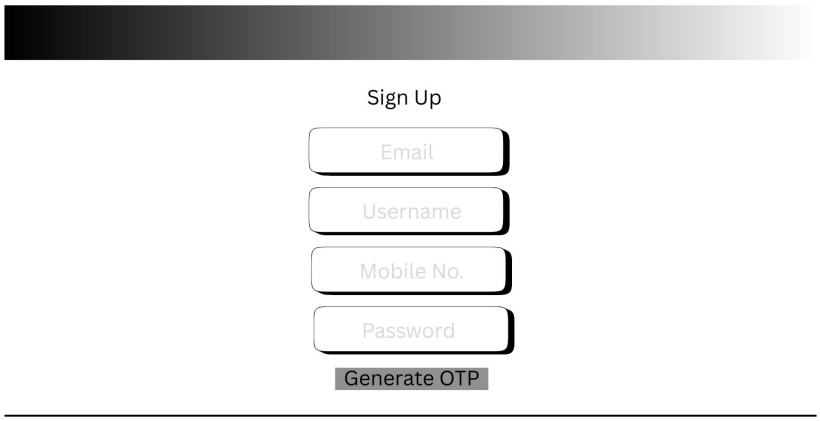
2. Sign In :

The Sign-In window will ask for the username and password of the user. The user can enter their username and password in the appropriate text fields and click on the login button at the bottom to sign in their account.



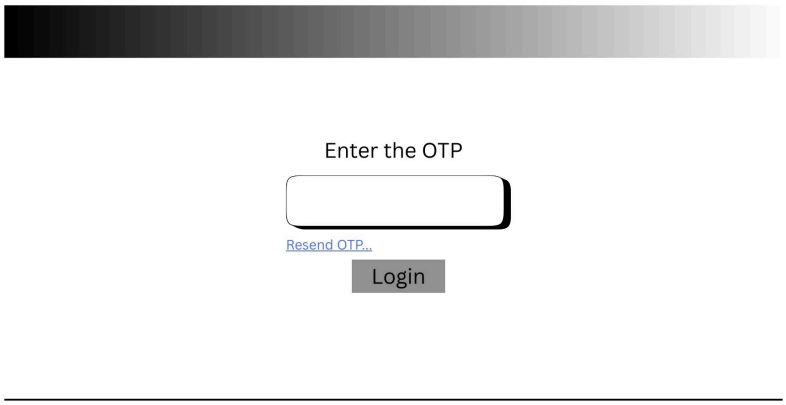
3. New User Registration :

The Sign-Up window will ask the new user to enter their email, username, mobile number and password in the respective input fields. After entering the required details, the user can click on the "Generate OTP" button to begin the account verification process.



4. OTP verification :

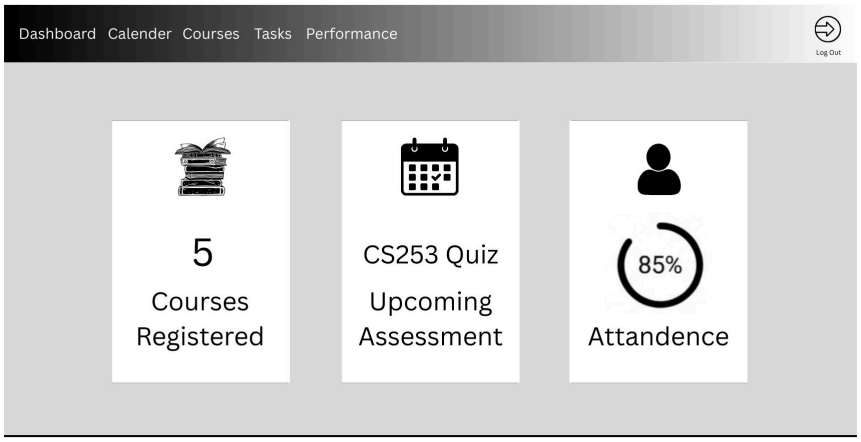
The OTP Verification window will prompt the user to enter the One-Time Password received on their registered contact. After entering the OTP, the user can click on the "Login" button to complete the verification. A "Resend OTP" option is also provided in case the user did not receive the code.



5. Interface :

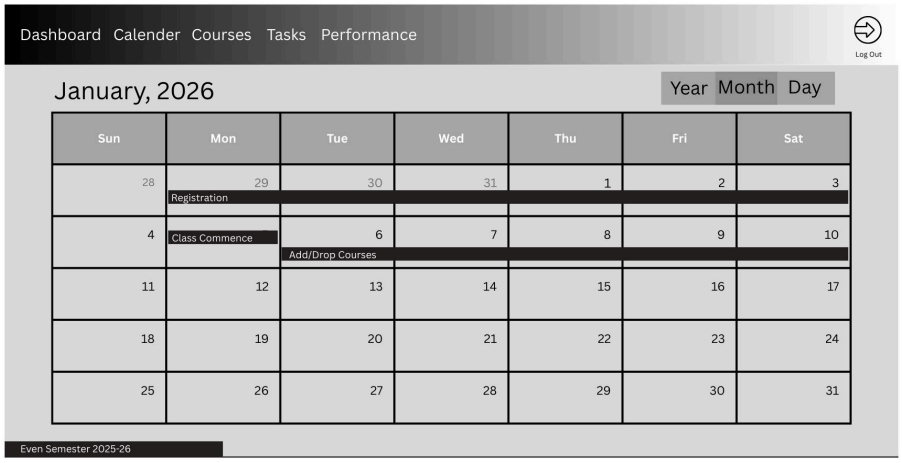
- ***Dashboard :***

The Dashboard provides an overview of the user's academic information. It displays key details such as the number of registered courses, upcoming assessments and attendance status. Navigation tabs allow the user to switch between different sections like Calendar, Courses, Tasks and Performance.



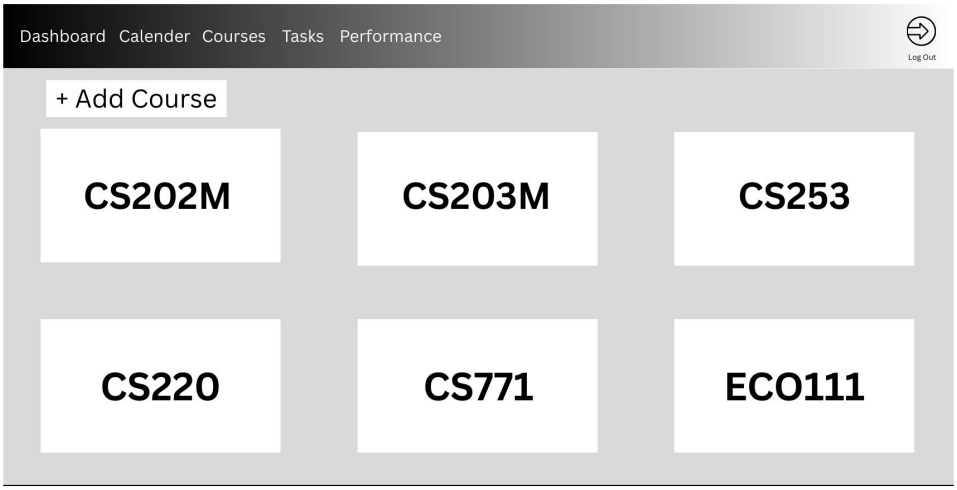
- ***Calender :***

The Calendar interface allows the user to view scheduled academic events such as quizzes, assessments and important deadlines. The user can switch between day, month and year views to easily track upcoming activities and manage their academic timeline.



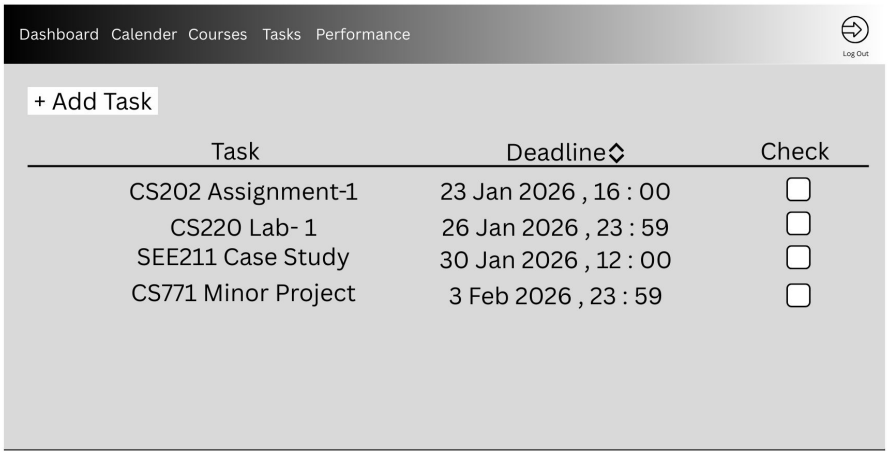
- ***Courses :***

The Courses section allows the user to view all the courses they are enrolled in. Each course is displayed as an individual card for quick access, and an "Add Course" option is available for registering new courses as needed.



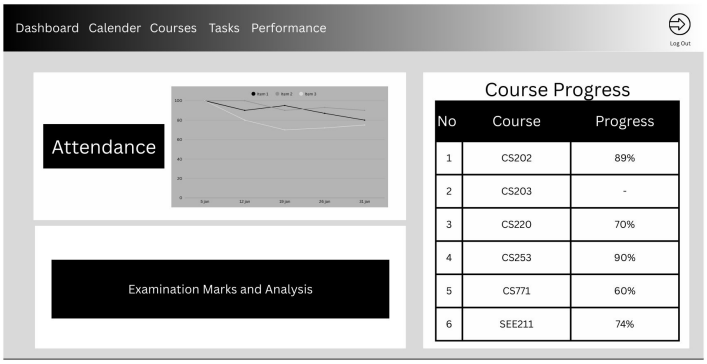
- ***Task :***

  The Task section displays a list of pending academic tasks along with their respective deadlines. Users can add new tasks using the "Add Task" option and mark completed tasks using the checkbox provided for each entry.



- ***Performance :***

The Performance section provides a detailed overview of the user's academic progress. It displays attendance information, course-wise performance metrics and examination analysis to help the user monitor their academic standing effectively.



3.1.2 Hardware Interfaces

There are no Hardware interfaces involved in the project. No additional hardware required.

3.1.3 Software Interfaces

There are no Software interfaces involved in the project. No additional software required.

3.2 Functional Requirements

Functional requirements capture the intended behavior of the system. This behavior may be expressed as services, tasks or functions the system is required to perform. This section is the direct continuation of section 2.2 where you have specified the general functional requirements. Here, you should list in detail the different product functions.

3.2.1 F1: User Account Management

- The system shall allow students to create an account using roll number/ phone number/ email.
- The system shall verify new user login using OTP Authentication.
- The system shall store and manage student academic data securely.

3.2.2 F2: Academic Manager

- The system shall allow students to record and track attendance for each course.
- The system shall allow students to store and manage academic resources for different courses.
- The system shall provide attendance shortage alerts, if any.

3.2.3 F3: Result and Performance Analysis

- The system shall allow students to view results of quizzes and tests.
- The system shall analyze academic performance using graphs.
- The system shall provide relative performance analysis across different subjects.

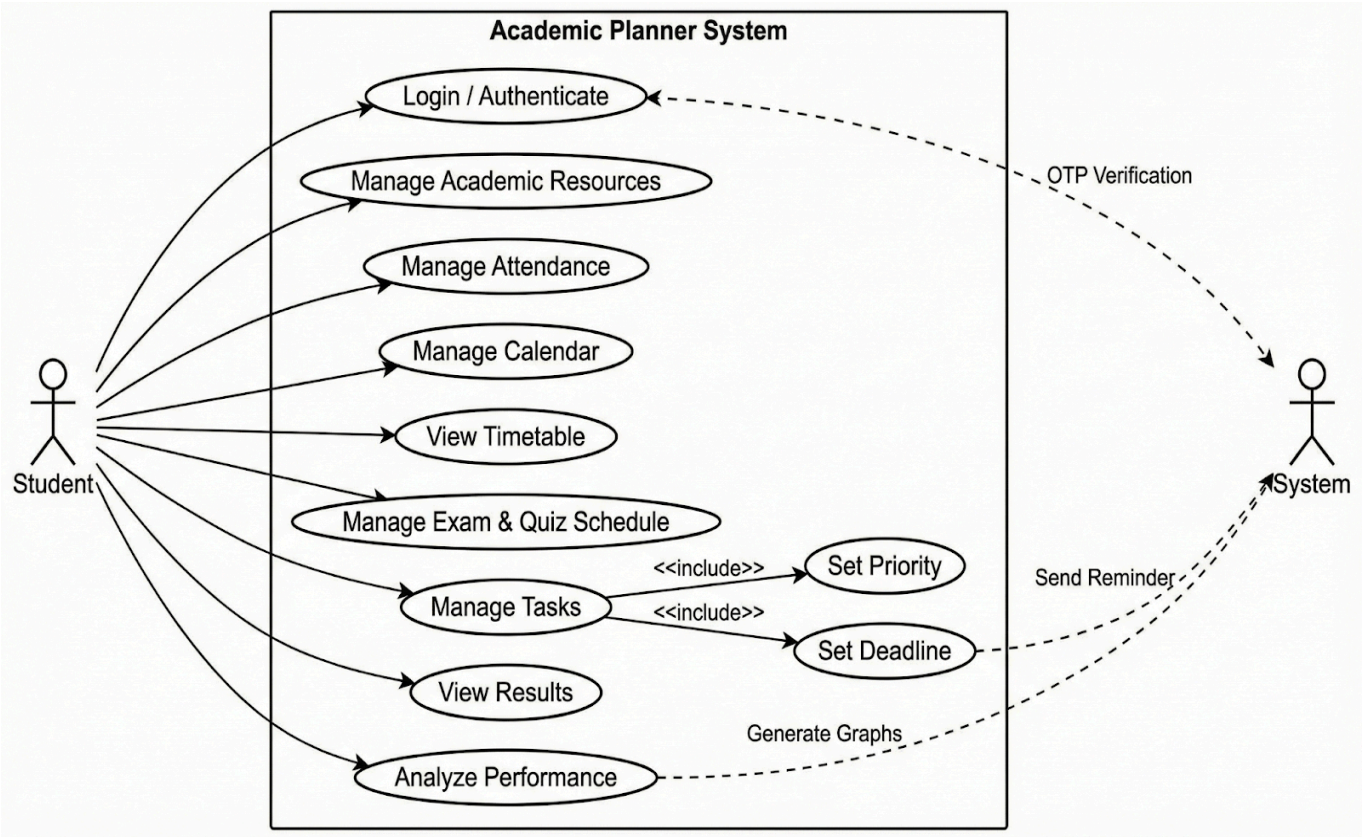
3.2.4 F4: Calendar Management

- The system shall allow students to view and manage exam schedules, quizzes, and venues.
- The system shall allow students to add submission deadlines and academic events to the calendar.
- The system shall provide a weekly timetable view.
- The system shall provide an upcoming exam, quiz and submission remainders.

3.2.5 F5: To-Do Management

- The system shall allow students to create daily and weekly tasks.
- The system shall allow students to prioritize tasks.
- The system shall send reminders to students before task deadlines.
- The system shall integrate long-term tasks with the calendar.

3.3 Use Case Model



3.3.1 Use Case U1: User Login and Authentication

Author – Deepak

Purpose - Allow students to securely access the application.

Requirements Traceability – F1

Priority - High

Preconditions - Student is registered in the system.

Post conditions - Student is logged in successfully.

Actors – Student, System

Exceptions - Invalid OTP, Incorrect credentials.

Includes - None

Notes/Issues - None

3.3.2 Use Case U2: Manage Academic Resources

Author – Deepak

Purpose - Allow students to manage course-related resources

Requirements Traceability – F2

Priority - Medium

Preconditions - Student is logged in.

Post conditions - Resources are added/updated successfully.

Actors – Student, System

Exceptions - Missing or invalid resource details.

Includes - None

Notes/Issues - None

3.3.3 Use Case U3: Manage Attendance

Author – Deepak

Purpose - Track attendance for each subject.

Requirements Traceability – F2

Priority - Medium

Preconditions - Student is logged in.

Post conditions - Attendance data is updated.

Actors – Student, System

Exceptions - Invalid attendance input.

Includes - None

Notes/Issues - None

3.3.4 Use Case U4: View Results and Performance Analysis

Author – Deepak

Purpose - Analyze academic performance.

Requirements Traceability – F3

Priority - Medium

Preconditions - Student is logged in and Student has test data available.

Post conditions - Performance graphs are displayed.

Actors – Student, System

Exceptions - No result data available.

Includes - None

Notes/Issues - None

3.3.5 Use Case U5: Manage Calendar Events

Author – Deepak

Purpose - Manage exams, quizzes, timetable and deadlines.

Requirements Traceability – F4

Priority - High

Preconditions - Student is logged in.

Post conditions - Calendar is updated with events.

Actors – Student, System

Exceptions - Event date conflict or missing details.

Includes - None

Notes/Issues - None

3.3.6 Use Case U6: Manage Tasks

Author – Deepak

Purpose - Create and organize tasks with reminders.

Requirements Traceability – F5

Priority - High

Preconditions - Student is logged in.

Post conditions - Tasks are created or updated.

Actors – Student, System

Exceptions - Invalid deadline or priority value.

Includes - None

Notes/Issues - None

4 Other Non-functional Requirements

4.1 Performance Requirements

This section defines the performance requirements of the Academic & Schedule Management System (ASMS) under normal, peak, and exceptional operating conditions. These requirements ensure that the system remains responsive, reliable, and usable for students accessing it from common devices such as mobile phones and laptops.

1. User Authentication Performance

Requirement

- User login requests shall be processed within **2 seconds** under normal conditions.
- OTP generation and delivery shall occur within **10 seconds**.
- OTP verification shall complete within **2 seconds** after submission.
- During peak usage, authentication response time shall not exceed **4 seconds**.

Rationale
Fast authentication is essential to ensure secure and convenient access to the system, especially when students frequently log in to check schedules or deadlines.

2. Academic Data Access Performance

Requirement

- Viewing academic data such as attendance records, course resources, or exam results shall load within **2–3 seconds**.
- Saving or updating academic data shall be completed within **2 seconds**.
- During peak usage, response time shall not exceed **5 seconds**.

Rationale
Students frequently access academic information, and delays could cause confusion regarding updated data.

3. Timetable and Academic Calendar Performance

Requirement

- Weekly timetables and academic calendars shall be displayed within **2 seconds**.
- Adding, editing, or deleting calendar events by professors or teaching assistants shall reflect in the system within **2-3 seconds**.
- Calendar data shall remain consistent across all user devices.

Rationale
Timetables and calendars are time-sensitive features, and immediate updates help students plan their academic activities accurately.

4. Task and To-Do Management Performance

Requirement

- Task creation, modification, prioritization, and deletion shall be processed within **1–2 seconds**.
- Task lists (daily, weekly, long-term) shall load within **2-3 seconds**.

Rationale
Task management is a core feature of ASMS. Quick interactions ensure smooth daily planning and productivity.

5. Reminder and Notification Timing (Real-Time Requirement)

Requirement

- Reminder notifications shall be delivered within **±1 minute** of the scheduled reminder time.
- Deadline reminders shall be sent at least **once before the deadline** as configured by the user.

Rationale
Reminders are time-critical features. Accurate and timely notifications are essential to help students meet academic deadlines.

6. Academic Performance Analysis and Graphs

Requirement

- Performance analysis graphs shall load within **3–4 seconds**.
- Graph rendering shall not cause the application to freeze or become unresponsive.
- Under peak load, graph loading time shall not exceed **6 seconds**.

Rationale
Performance analytics involve data processing and visualization, which may take slightly longer. However, responsiveness must still be maintained for usability.

7. Concurrent User Performance (Peak Load)

Requirement

- The system shall support **at least concurrent users** without significant degradation in response time.

- Under peak usage (e.g., before exams), response times may increase but shall not exceed **5 seconds** for any user action.

Rationale
Peak usage is expected during academic deadlines and exam periods. The system must remain usable during such times.

8. Availability and Reliability Performance

Requirement

- The system shall be available **99% of the time**, excluding scheduled maintenance.
- In the event of a failure, the system shall recover without data loss.

Rationale
High availability ensures students can rely on the system for daily academic planning without unexpected downtime.

9. Network Performance Constraints

Requirement

- The system shall be accessible **only through the campus Wi-Fi network or via the campus VPN**.
- On slower networks, essential features shall still load, even if response times increase slightly.

Rationale
Restricting access to campus Wi-Fi or the VPN improves security and ensures a controlled network environment. Accounting for VPN latency allows developers to design efficient networking, caching, and timeout mechanisms while maintaining acceptable performance for students accessing the system remotely. Students may access ASMS from various network conditions, including low-bandwidth environments.

10. Resource Utilization

Requirement

- The application shall not consume excessive device memory or battery power.
- Background processes such as notifications shall be optimized for efficiency.

Rationale
Efficient resource usage ensures smooth operation on mobile devices and improves user satisfaction.

4.2 Safety and Security Requirements

The system stores personal and academic information of students, therefore basic safety and security measures are required to protect the data and prevent misuse of the application.

Safety Requirements :

The system shall ensure that student data is not accidentally lost or deleted.

The system shall ask for user confirmation before performing actions that may remove important academic or task related data.

The system shall handle incorrect inputs in a safe manner without crashing.

Security Requirements :

Users shall be required to log in using valid credentials such as roll number along with email or phone number.

***OTP** verification shall be used during login to improve security.*

*User login details **shall not be stored in plain text**.*

Only authorized users shall be able to access academic records, schedules, and task information.

The system shall not display sensitive internal details in case of errors.

4.3 Software Quality Attributes

4.3.1 Availability :

This web application will be available and accessible to every user across the campus.

4.3.2 Usability :

*This web application **UI** shall automatically resize and rearrange elements to fit screen widths ranging from **320px** (mobile) to **1920px** (Desktop) without horizontal scrolling. The **UI** of this application will be user friendly and easy to use.*

4.3.3 Reliability :

*The system must ensure that **student logs** (attendance and grades) are never lost due to browser closures or minor connectivity issues. The application shall utilize Local Storage to save data instantly as the user types to solve this. The system shall achieve an **Uptime** (Continuous operational time) of 99.5% during the active semester.*

4.3.4 Privacy :

*The system shall ensure Strict Data Isolation. The **API** shall reject any request to view or modify data that does not match the currently authenticated user's token. All personal; Data transmission must occur over and encrypted connection.*

Appendix A – Data Dictionary

Will be updated later.

Appendix B - Group Log

Since the beginning of the project, our entire team has been very enthusiastic. We have formed a Whatsapp group for effective communication.

Meeting minutes	Agenda
11 Jan 2026	Discussed different ideas for the project and finalized one idea for a student assistant app.